

What Does “N× Fewer Tokens” Measure? Baselines, Sampling, and Output Form in Repository-Navigation Benchmarks

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Abstract—Code-graph tools for coding agents are promoted with large token-reduction ratios. We ask how much such a ratio depends on how it is measured. We contribute NAVBENCH, a model-free benchmark whose labels never come from an evaluated tool: seeded fixtures with known call sites, 17 public Python and TypeScript repositories with independently sampled targets, and calls observed while running the test suites. We evaluate two Tree-sitter graph tools, ripgrep, and two language servers. Re-applying our own prototype’s earlier benchmark definition reproduces its headline: graph answers are 85× smaller than the whole files they replace. The same answers are only 3.0× smaller than `rg -w` output, and up to 2.2× larger than compact language-server output. Sampling targets from the prototype’s own edges raises its answer rate from 44–54% to 99–100%; 32–42 points remain after adjusting for target composition. For direct callers, the most complete graph tool used as many tokens as ripgrep on targets both answered completely (ratio 1.03, 95% CI 0.94–1.13), but with far fewer wrong candidates (precision 0.98 vs. 0.35). For depth-2 callers it needed 4.5× fewer tokens (0.22, 0.20–0.24). In four large repositories, ripgrep’s relative payload did not grow (3.2×), while graph-tool recall on observed calls fell to 0.42 (ripgrep 0.78). We release the harness and data, and propose that efficiency claims report baseline, sampling frame, output form, completeness, and precision together.

I. INTRODUCTION

Coding agents spend much of their context budget locating code [13]. Graph-based navigation tools, exposed to the model as MCP tools, promise to answer structural questions such as “who calls *X*?” from a pre-built index [1], [2], [3], [4], and are often promoted with a token ratio. A ratio has three hidden parameters that are not properties of the tool: the *baseline* (whole files, grep, or a language server), the *query sample* (often symbols the graph already connects), and the *output form* (names, locations, or source). Our starting point was a research prototype by the authors, G1, whose own benchmark reported 85.8× fewer tokens than reading the referencing files, on graph-selected symbols; we evaluate G1 as one tool among others and report its failures.

We ask: **Q1** How does the baseline change measured ratios? **Q2** How does sampling targets independently of the graph change measured coverage? **Q3** At matched completeness, with labels no tool produced, what do tools cost, for direct and for multi-hop callers?

Contributions: (1) NAVBENCH, an open harness with three independent label layers, automatic validity gates, and output-form control; (2) evidence that one set of graph outputs measures 0.45× to 139× depending on baseline and form; (3) a

quantification of graph-conditioned sampling bias, adjusted for target composition; (4) matched-completeness costs showing no saving for direct callers but a 4.5× saving for depth-2 callers, with recall on runtime-observed calls; and (5) a large-repository falsification test.

II. BENCHMARK DESIGN

Tasks. T1 (primary): given a declaration, return its direct call sites (call expressions whose callee statically binds to it). T2: given a call site, return the declaration. T3 (added after freezing; fixtures only): callers up to depth 2.

Layer A: seeded fixtures. A generator emits small executable Python and TypeScript repositories with one instance of each pattern: direct calls; calls through an import alias, a module or namespace attribute, a re-export, a typed or super receiver; constructor and module-level calls; a same-named method on another class; a shadowing local; callbacks; string dispatch; decorators or arrow functions; and comment and string distractors. Half use common, reused names. A manifest lists every declaration and call edge; Python fixtures execute and TypeScript fixtures compile under `-strict`. A gate requires the runtime-observed Python calls to equal the manifest; it caught three manifest omissions before any held-out data existed. We use 4+4 calibration and 16+16 held-out fixtures (160 targets with explicit call sites), plus a second held-out split from fresh seeds.

Layer B: natural repositories. 9 Python and 8 TS/JS libraries, chosen by fixed criteria before any result. Declarations are enumerated by tools sharing no code with any graph tool: Python `ast` and the TypeScript compiler API. *S-ind* (661 targets) is a stratified random sample of non-test declarations (strata: kind × language-server call fan-out; inclusion weights recorded). *S-cond* (331 targets) is drawn from declarations with ≥1 incoming G1 edge, the original benchmark’s rule. Layer B supports availability, size, and agreement, not accuracy.

Layer C: runtime-observed calls. Each Python test suite runs under `sys.monitoring`; a call counts only when the callee’s entry is confirmed. Following dynamic-baseline evaluation of call graphs [12], we report recall on observed calls; unobserved static results are not false positives.

Tools. `rg -w` with 0 and 3 context lines; pyright 1.1.414 and the TypeScript 5.9.3 language service (references; native form: LSP Location JSON); G1 `find_references` and

TABLE I

Q1: BASELINE/GRAPH PAYLOAD RATIO (REPO-MACRO GEOMETRIC MEAN, S-IND; EACH ROW OVER THAT TOOL’S NON-EMPTY ANSWERS, N = TARGETS).

Graph tool	n	whole files	rg-w	LSP JSON	LSP loc.
G1 find_references	358	139.3	6.01	3.10	0.88
G1 lookup+deps	290	85.6	3.27	2.46	0.54
codebase-memory current	395	48.0	2.92	1.92	0.45
codebase-memory v0.5.x	413	60.8	3.54	1.46	0.46

its original `lookup_symbol+get_dependencies` pair; codebase-memory [4] at current main (bf93f0b7) and at v0.5.5, the release its paper evaluates. v0.5.5 and v0.5.7 agree on every fixture query and differ on 5.8% of natural queries, no more than v0.5.7 differs from its own re-run (5.3%), so we report them as one row (“v0.5.x”). Tools are queried through their own interfaces (MCP sessions for codebase-memory); a qualified name is sent only after a tool itself reports ambiguity.

Scoring and statistics. Facts are mapped to call sites, enclosing callers, and files; an answer is *complete* when caller recall is 1. Tokens (cl100k; o200k changes `rg-w/graph` ratios by ≤ 0.06) are counted on native output, a common one-fact-per-line form, and a source-enriched form, with an automatic check that every form carries the same facts. Estimates are repository-macro means with 95% hierarchical-bootstrap intervals (repositories, then targets). The primary contrast, the token ratio to `rg-w` on targets both arms answer completely, and a $1.5\times$ margin were frozen before held-out data.

III. RESULTS

A. Q1: the baseline determines the ratio

Re-applying G1’s original definition (graph-conditioned targets, whole files selected by the graph, `lookup+deps` output) to the public repositories gives $85.3\times$ (95% CI 57.2–126.9; 328 targets), matching the $85.8\times$ originally reported. This reproduces G1’s own measurement only; it does not test codebase-memory’s agent-level $10\times$ result [4], which measures a different workflow. Against `rg-w` the same outputs are $3.0\times$ smaller (2.4–4.0), and against the language server printed one location per line they are larger (Table I, Fig. 1). On the 226 targets where every graph tool answered, the ordering is unchanged (`rg-w` 2.5–4.7 $\times$; LSP location form 0.45–0.86 $\times$). The `rg-w/graph` ratio grows with the number of occurrences of the target’s name (exploratory elasticity 0.77, CI 0.69–0.87): lexical search is costly mainly for common names.

B. Q2: graph-conditioned sampling overstates coverage

S-cond also changes target composition: 9% of its targets but 35% of S-ind’s have no call site according to the language server, and for those an empty answer is correct. We therefore post-stratify S-cond to S-ind’s distribution over repository $\times$ kind $\times$ fan-out bucket (0, 1–2, 3–9, ≥ 10). The adjustment works as a control: the language server, on which S-cond is not conditioned, shows no adjusted difference (Table II). For G1, the conditioning tool, 32–42 points of answer rate remain; its observed-call recall gap shrinks to +0.13 (CI reaching 0). For

TABLE II

Q2: S-COND $\rightarrow$ S-IND. NON-EMPTY ANSWERS (%), ADJUSTED S-COND–S-IND DIFFERENCE [95% CI], AND CALLER RECALL ON RUNTIME-OBSERVED CALLS.

Tool	non-empty %	adjusted Δ (pts)	obs.-call recall
G1 find_references	100 $\rightarrow$ 54	+32 [21, 41]	0.85 $\rightarrow$ 0.70
G1 lookup+deps	99 $\rightarrow$ 44	+42 [29, 53]	0.81 $\rightarrow$ 0.63
codebase-memory current	88 $\rightarrow$ 60	+9 [2, 17]	0.79 $\rightarrow$ 0.66
codebase-memory v0.5.x	84 $\rightarrow$ 62	+8 [0, 15]	0.72 $\rightarrow$ 0.60
language server	97 $\rightarrow$ 83	−1 [−5, 3]	0.84 $\rightarrow$ 0.77
rg-w	100 $\rightarrow$ 100	0	0.97 $\rightarrow$ 0.95

codebase-memory the adjusted gap is 8–9 points: conditioning inflates mainly the coverage of the tool whose edges define the sample. With the recorded inclusion weights, the S-ind answer rates fall further (codebase-memory current 60% $\rightarrow$ 54%). G1 misses 14% of independently sampled declarations; in five of eight TS/JS repositories its coverage is only 47–68% (unindexed object-literal functions, collapsed same-named nested declarations).

C. Q3: cost at matched completeness

Direct callers (T1). Table III and Fig. 2 report the held-out fixtures. The primary contrast is the geometric-mean token ratio to `rg-w` over targets both arms answer completely; these subsets differ by arm and are conditioned on success. A saving counts as meaningful only if the whole interval lies below $1/1.5 \approx 0.667$: G1 find_references 0.35 [0.33, 0.37] (n=64) and codebase-memory v0.5.x 0.58 [0.56, 0.62] (n=79) save, but are complete on only 40% and 49% of targets; G1 lookup+deps 0.65 [0.62, 0.672] does not qualify; codebase-memory current, the only graph tool as complete as the baselines, gives 1.03 [0.94, 1.13] (n=127): no evidence of a saving (not an equivalence test); language-server JSON costs 1.75 [1.65, 1.85]. Without conditioning on completeness codebase-memory current’s ratio is 1.01 [0.93, 1.09], and the fresh-seed split reproduces every contrast (codebase-memory current 1.02 [0.94, 1.12]).

Matched completeness is not matched quality: codebase-memory current returns almost no wrong callers (precision 0.98 vs. 0.35; F1 0.95 vs. 0.50). The fixtures plant same-named methods, shadowing locals, and textual mentions, so they are deliberately hard for lexical search. Form matters again: in the common one-fact-per-line form every graph tool needs only 0.24–0.28 of `rg-w`’s tokens on the same targets, so codebase-memory’s native parity comes from its verbose tree output (160 vs. 19 tokens); graph lines name callers, whereas `rg-w` lines list every occurrence.

Depth-2 callers (T3, exploratory). Here the graph tool pays off: codebase-memory current is complete on 89% of targets at 181 tokens, `rg-w` on 80% at 654, and on targets both complete it needs 0.22 [0.20, 0.24] of `rg-w`’s tokens (n=127) and 0.42 [0.38, 0.47] of the language server’s. The `rg-w` strategy re-greps each caller and reads file outlines, a deliberately complete lexical baseline.

Failures by pattern. `rg-w` and pyright miss only calls through import aliases. Of G1’s 240 missed callers

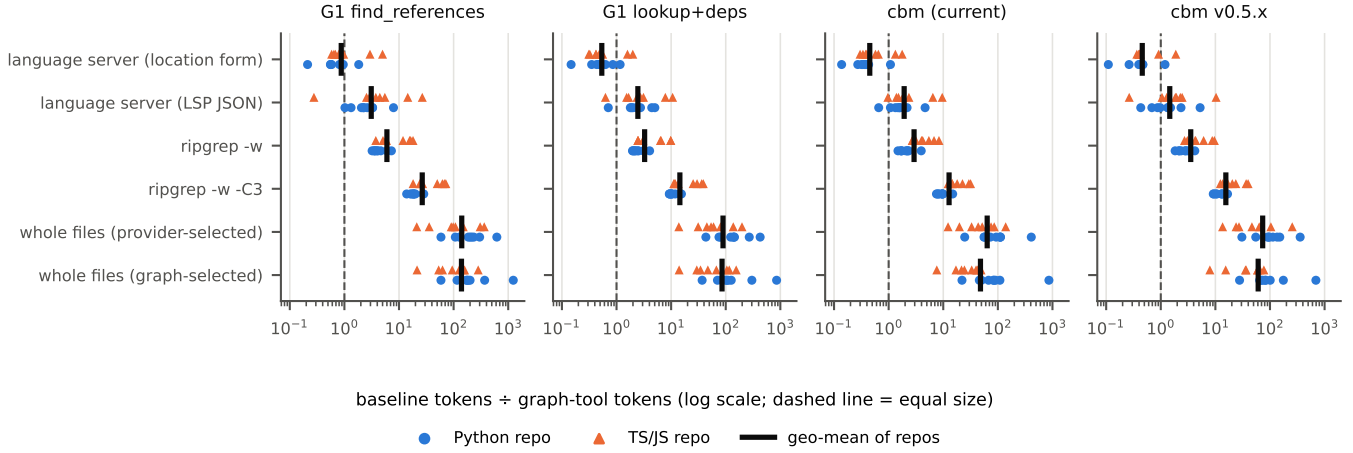


Fig. 1. Q1. Each baseline’s payload divided by each graph tool’s native output, for the same queries (S-ind, non-empty graph answers). Points: repository geometric means; bars: mean over repositories. Baseline and output form move the ratio by two orders of magnitude.

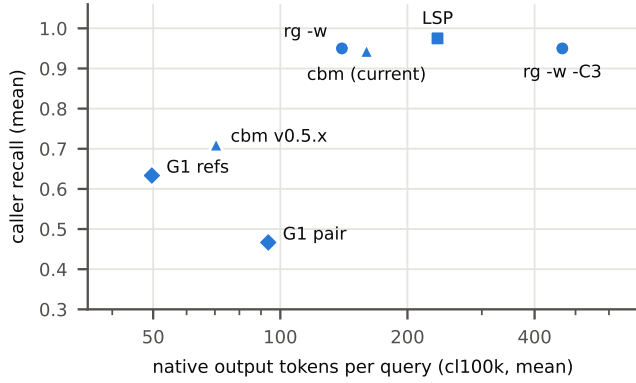


Fig. 2. Q3, T1, held-out fixtures: native tokens vs. caller recall (repository means; G1 pair = lookup+deps). Small outputs coincide with low recall.

TABLE III

Q3: HELD-OUT FIXTURES (160 TARGETS). CALLER RECALL, PRECISION, COMPLETE ANSWERS, AND MEAN NATIVE TOKENS FOR DIRECT CALLERS (T1); COMPLETE ANSWERS AND TOKENS FOR DEPTH-2 CALLERS (T3).

Tool	T1				T3	
	rec.	prec.	compl.%	tok.	compl.%	tok.
rg -w	0.95	0.35	80.0	140	80.0	654
language server	0.97	0.64	90.0	236	90.0	402
G1 find_references	0.63	0.68	40.0	50	30.0	102
G1 lookup+deps	0.47	1.00	40.0	94	—	—
codebase-memory current	0.94	0.98	89.4	160	89.4	181
codebase-memory v0.5.x	0.71	0.90	49.4	70	—	—

(lookup+deps), 96 resolve to a shadowing local, 32 are module-level calls, and 112 are unrecorded relations (re-exports, TS constructors, typed receivers). Ambiguous names lower G1 find_references recall from 0.73 to 0.53 but leave codebase-memory current unaffected. In T2 every tool found every definition; only the language server returned exactly one.

D. Layer C: recall on runtime-observed calls

On 323 Python targets with observed calls (both samples, each target once), caller recall was `rg -w` 0.95 [0.92, 0.98], language server 0.79 [0.64, 0.90], G1 0.75 and 0.69, and codebase-memory 0.70 (current) and 0.64 (v0.5.x). The gap is not explained by unindexed tests: on callers in library code the graph tools reach 0.63–0.76, against 0.94 for `rg -w` and 0.86 for the language server. Pyright’s misses concentrate in repositories that ship `.pyi` stubs, to which it binds public-API calls. The graph tools’ recall is in the range reported for static Python call graphs [9].

E. Falsification test: large repositories

Because the `rg -w/graph` ratio grows with name frequency, graph tools should gain in large repositories. We ran the frozen protocol on four large repositories chosen by fixed criteria before any result (networkx, SQLAlchemy, TypeORM, NestJS; 7.5k–41k declarations; 158 S-ind targets). The prediction fails. `rg -w`’s payload relative to codebase-memory current is $3.2\times$ (2.4–4.5; 2.8–3.6 per repository), as in the main corpus ($2.9\times$), because independently sampled declarations are no more ambiguous there (median 12.5–18 name occurrences, vs. 15). Coverage worsens instead: codebase-memory current answers 50–67% of S-ind targets, and on runtime-observed calls its caller recall is 0.42, against 0.78 for `rg -w` and 0.67 for the language server (29 targets; the frozen tracing budget covered only 9% and 2% of the two Python suites). G1 stops at a fixed symbol budget, indexes 23–62% of declarations, and answers 10–44% of targets.

IV. RECOMMENDATIONS

- Report several baselines: one set of graph outputs measured $0.45\times$ to $139\times$.
- Sample targets independently of the tool under test; sampling from G1’s own edges added 32–42 points of answer rate after adjustment.

- Count tokens in a common output form and state what it contains.
- Report completeness and precision next to tokens, and separate single-hop from multi-hop queries: the same tool saved nothing on the first and $4.5\times$ on the second.
- Do not equate payload with agent cost [13], [14]; agent-level studies [4], [5], [6] measure something different.

V. THREATS TO VALIDITY

Construct. Payload tokens are not agent cost. Labels encode one static semantics; caller-level scoring credits any location inside a true caller, but crediting only gold call sites changes recall by ≤ 0.01 .

Internal. G1 is the authors’ tool; every tool runs through its own interface under one frozen configuration, and G1’s failures are reported in full. Post-freeze changes, each released with before/after results: a pyright readiness wait (layer-C recall $0.73 \rightarrow 0.79$; Python targets were re-drawn with the same seed); a scoring fix for the 76 targets drawn into both samples (S-ind numbers changed, conclusions did not); codebase-memory v0.5.5; T3, the fresh split, the large-repository extension, and the robustness analyses (weights, common subset, strict credit, post-stratification). codebase-memory v0.5.x answers for same-named declarations vary between runs (5% of natural queries; current 0.4%), and G1’s answers vary when truncated at its output caps.

External. Fixtures are cleaner than real code; repositories are a purposive selection in two languages; Layer C covers only exercised code; there is no human audit of natural-code accuracy.

VI. RELATED WORK

Code graphs improve repository-level tasks for agents [1], [2], [3]. Agent-level studies report token savings under different setups: Codebase-Memory about $10\times$ fewer tokens than a grep-and-read agent at lower answer quality (83% vs. 92%) [4]; CodeNib 50–87% fewer tokens on SWE-bench issue queries under a quality-preserving margin [6]; Xu finds that language servers seldom save agent tokens and studies when they do [5]. Retrieval benchmarks score which files or snippets an agent should read [7], [8]; call-graph benchmarks score edges, with micro-benchmarks, LLMs, and dynamic baselines [9], [10], [11], [12]. We instead score structural answers as context: their tokens at matched completeness, the sampling frame, and the output form, without a model.

VII. DATA AVAILABILITY

Harness, fixture generator, pinned corpus commits, raw per-query outputs, traces, analyses (including the large-repository extension), frozen protocol, and a one-command reproduction script: <https://anonymous.4open.science/r/XXXX> (anonymized).

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